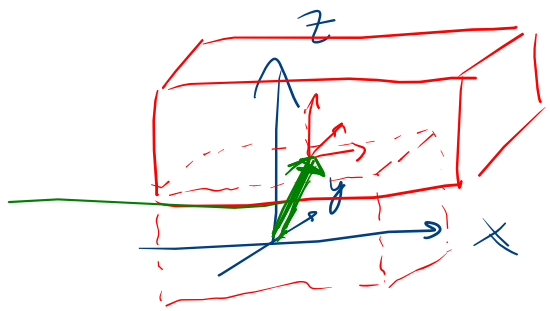
spatialmath and Swift sim
derivatives (time and partial)
Lie derivatives
gradients
Jacobians
- through rigid body transformations
Twists

Note: in URDFs, you should think of the <origin> tag as setting the translation part of a transformation

origin 1,0,2

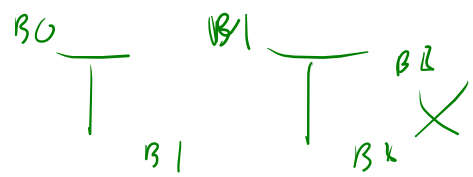


origin 0,0,0

spatialmath
Python module for rigid body transformations and other related implementations

The class SE3 implements rigid body transfomations (4x4 homogeneous transformations)

SE3 provides all elementary transformations (translations and rotation around the principal axes)



(Python) Robotics Toolbox (by Peter Corke

Module to implement common operations in robotics

Main concept: ETS (Elementary Transformation Sequence) Robot
A class that contains links, joints, and methods to perform simple operations on the robot

roboticstoolbox.models.catalog() Shows the list of robots that are built-in the toolbox

ERobot: panda (by Franka Emika), 7 joints (RRRRRRR), 1 gripper, geometry, collision

link	link	joint	parent	ETS: parent to link
0	panda_link0		BASE	SE3()
1	panda_link1	0	panda_link0	SE3(0, 0, 0.333) ⊕ Rz(q0)
2	panda_link2	1	panda_link1	SE3(-90°, -0°, 0°) ⊕ Rz(q1)
3	panda_link3	2	panda_link2	SE3(0, -0.316, 0; 90°, -0°, 0°) ⊕ Rz(q2)
4	panda_link4	3	panda_link3	SE3(0.0825, 0, 0; 90°, -0°, 0°) ⊕ Rz(q3)
5	panda_link5	4	panda_link4	SE3(-0.0825, 0.384, 0; -90°, -0°, 0°) ⊕ Rz(q4)
6	panda_link6	5	panda_link5	SE3(90°, -0°, 0°) ⊕ Rz(q5)
7	panda_link7	6	panda_link6	SE3(0.088, 0, 0; 90°, -0°, 0°) ⊕ Rz(q6)
8	@panda_link8		panda_link7	SE3(0, 0, 0.107)

end effector

link names

joint numbers

transformations given by joints

Swift simulator

Note: on pip, this module is called swift-sim, not ~~swift~~

Light-weight visualization for the Robotics Toolbox. Uses JavaScript

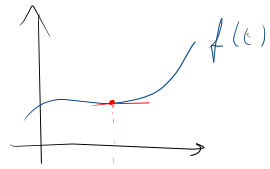
Centered around the concept of
1) Creating an environment
2) Add robots, obstacles
3) A loop that
   3a) Updates robots/obstacles
   3b) Advances a simulation "step"

```
env=swift.Swift()
env.launch(realtime = True)
env.add(robot)
while True :
    robot.q = [0 ,0]
    env.step (0.05)
```

how much the simulated time should advance

Derivatives

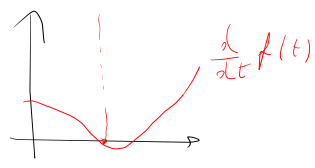
$f(t) \in \mathbb{R} \longrightarrow \dot{f} = \frac{\partial}{\partial t} f(t) \in \mathbb{R}$

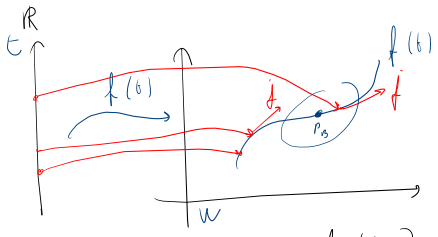


$f(t) = \begin{bmatrix} f_1(t) \\ f_2(t) \\ f_3(t) \end{bmatrix} \longrightarrow \dot{f} = \frac{\partial}{\partial t} f(t) = \begin{bmatrix} \frac{\partial}{\partial t} f_1(t) \\ \frac{\partial}{\partial t} f_2(t) \\ \frac{\partial}{\partial t} f_3(t) \end{bmatrix}$

parametric curve

vector = tangent of the curve
at that value of t
= velocity





if f is the translation part of a rigid body transformations

-> $\dot{f}$ is the linear velocity of the (in the world frame)

$\dot{p}_B$

$f(t) = \begin{bmatrix} f_{11}(t) & f_{12}(t) \\ f_{21}(t) & f_{22}(t) \end{bmatrix} \longrightarrow \dot{f} = \begin{bmatrix} \frac{\partial}{\partial t} f_{11}(t) & \frac{\partial}{\partial t} f_{12}(t) \\ \frac{\partial}{\partial t} f_{21}(t) & \frac{\partial}{\partial t} f_{22}(t) \end{bmatrix}$

$R(t) = \begin{bmatrix} \cos q(t) & -\sin q(t) \\ \sin q(t) & \cos q(t) \end{bmatrix} \longrightarrow \dot{R} = \begin{bmatrix} \frac{\partial}{\partial t} \cos q(t) & \frac{\partial}{\partial t} -\sin q(t) \\ \frac{\partial}{\partial t} \sin q(t) & \frac{\partial}{\partial t} \cos q(t) \end{bmatrix} = \begin{bmatrix} -\dot{q} \sin(q) & -\dot{q} \cos(q) \\ \dot{q} \cos(q) & -\dot{q} \sin(q) \end{bmatrix}$

angular velocity

$= \begin{bmatrix} \cos q(t) & -\sin q(t) \\ \sin q(t) & \cos q(t) \end{bmatrix} \begin{bmatrix} 0 & -\dot{q} \\ \dot{q} & 0 \end{bmatrix}$

R(t) $\quad\quad \hat{\omega}(t)$

skew-symmetric

$\begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} 0 & \omega_3(t) & -\omega_2(t) \\ -\omega_3(t) & 0 & \omega_1(t) \\ \omega_2(t) & -\omega_1(t) & 0 \end{bmatrix}$

these values can be interpreted as the instantaneous angular velocities along the three main axes

$\begin{bmatrix} \omega_1(t) \\ \omega_2(t) \\ \omega_3(t) \end{bmatrix}^\wedge$

skew symmetric matrix appears on the right, so is in "body" coordinates

(you could make a similar construction with skew symmetric times R(t), in that case we end up with the velocity in "world" coordinates)

$\dot{R}_B(t) = \begin{bmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix}$

A matrix is said to be skew-symmetric if $A = -A^T$

this implies zero diagonal

$\begin{bmatrix} 0 & -\omega_1 \\ \omega_1 & 0 \end{bmatrix}$ $\quad\quad \begin{bmatrix} 0 & \omega_3 & -\omega_2 \\ -\omega_3 & 0 & \omega_1 \\ \omega_2 & -\omega_1 & 0 \end{bmatrix}$

1 independent parameter

3 independent parameters

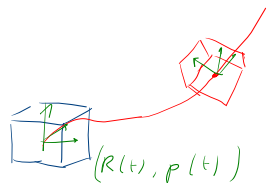
$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$

Rigid body poses (+ homogeneous transformations)

$\frac{\partial}{\partial t}\left({}^W R_B(t), {}^W p_B \right) \longrightarrow \left({}^W \dot{R}_B(t), {}^W \dot{p}_B \right)$

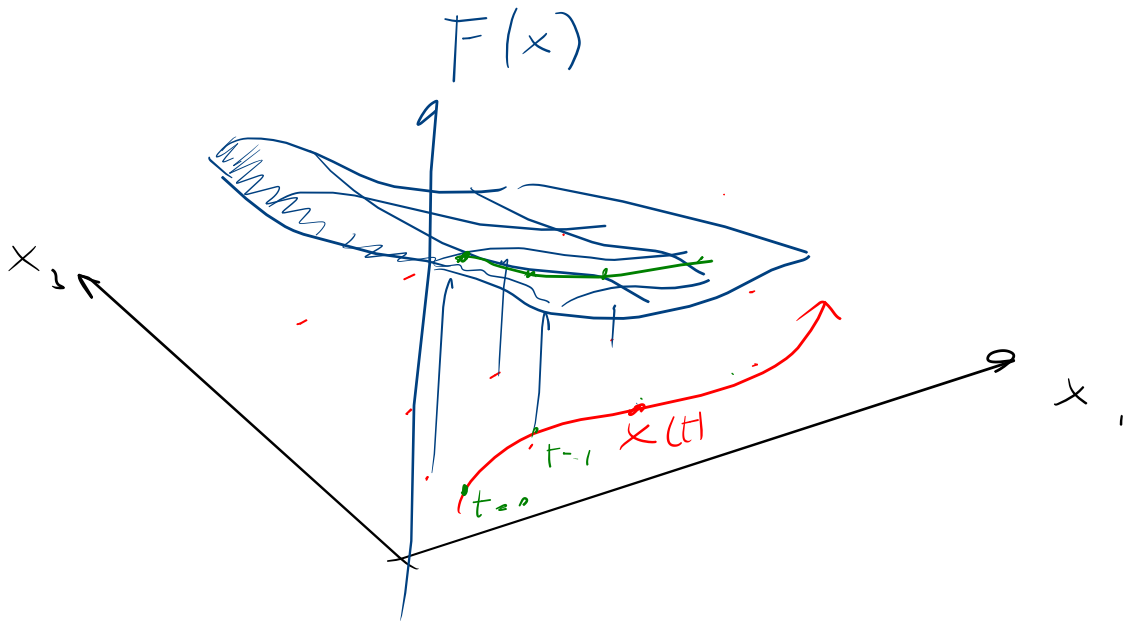
$\begin{bmatrix} {}^V R_B & {}^V p_B \\ 0^T & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} {}^W \dot{R}_B & {}^W \dot{p}_B \\ {}^V R_B \hat{\omega} & 0 \\ 0^T & 0 \end{bmatrix}$

Twist $\quad \left(\omega, \dot{p} \right)^\wedge$



Lie derivatives

$F(x) : \mathbb{R}^n \to \mathbb{R}$ scalar field: input points, output values



$x(t) : \mathbb{R} \to \mathbb{R}^n$

$$\mathcal{L}_{\dot{x}} F = \frac{\partial}{\partial t} F(x(t))$$

$F(x(t)) : \mathbb{R} \to \mathbb{R}$

